

Name: _____

Date: _____



ratio & proportion - level 3



equivalent ratios

1 fill in the missing number

Make each pair of ratios equivalent.

$2 : 3 = 4 : \boxed{}$

$3 : 5 = 9 : \boxed{}$

$4 : 7 = \boxed{} : 21$

$5 : 8 = 15 : \boxed{}$

$6 : 9 = 2 : \boxed{}$

$7 : 4 = \boxed{} : 12$

$3 : 2 = 12 : \boxed{}$

$8 : 10 = 4 : \boxed{}$

2 match equivalent ratios

Draw a line to match each ratio to an equivalent ratio.

1 : 2



10 : 12

2 : 3



8 : 10

3 : 4



6 : 8

4 : 5



4 : 6

5 : 6



2 : 4

3 scale up and down

Complete each ratio chain.

2 : 5



3 : 4



5 : 7



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ratio & proportion - level 5



proportion problems

1 solve the proportion problems

Use equivalent ratios or scaling to solve each problem.

1 The ratio of girls to boys is 3 : 2. If there are 15 girls, how many boys are there?

answer:

2 A map uses a scale of 2 cm : 5 km. A route is 6 cm on the map. How far is it really?

answer:

km

3 Red marbles : blue marbles = 2 : 3. If there are 18 blue marbles, how many red marbles?

answer:

4 Juice : water = 1 : 4. If you use 250 ml of juice, how much water is needed?

answer:

ml

5 Books : pencils = 5 : 3. If there are 20 books, how many pencils are there?

answer:

6 A recipe uses 3 cups flour for 2 cups milk. How much milk is needed for 12 cups flour?

answer:

cups

2 make your own

Create a proportion problem using the ratio 2 : 5.

Write your problem:

Show your working:

answer:

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ratio & proportion - level 4



scaling

1 scale a recipe

A small recipe uses flour : sugar = 2 : 3.

Complete the table when the recipe is scaled.

ingredient	x1	x2	x3	x5
flour	2			
sugar	3			

If the recipe is $\times 4$, flour =

sugar =

2 scale quantities

Use the scale factor to complete each line.

3 pens cost £6. 6 pens cost £

4 bottles hold 8 L. 12 bottles hold

L

5 tickets cost £15. 10 tickets cost £

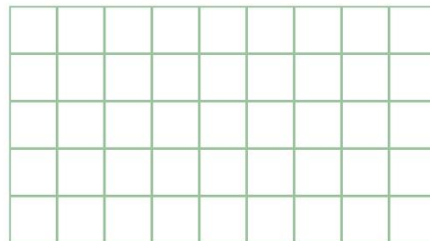
2 boxes contain 12 balls. 8 boxes contain

balls

3 draw the scaled shape

A rectangle is 3 squares wide and 2 squares tall.

Draw it at scale factor $\times 2$ in the grid.



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ratio & proportion - level 1



visual ratios

1 write the ratio

Count the pictures. Write each ratio in the boxes.

red circles to blue circles



ratio: :

yellow circles to purple circles



ratio: :

green circles to pink squares



ratio: :

orange circles to blue squares



ratio: :

purple circles to green circles



ratio: :

2 circle the matching ratio

Circle the ratio that matches each picture group.



3:2

2:3

5:2



2:4

4:2

1:4



5:1

1:5

4:1

3 draw your own ratio

Draw objects to show each ratio.

2 : 3

4 : 1

3 : 5

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ratio & proportion - level 2



simplify ratios

1 simplify the ratios

Divide both parts by the same number.

$6 : 9 = \boxed{} : \boxed{}$

$8 : 12 = \boxed{} : \boxed{}$

$10 : 15 = \boxed{} : \boxed{}$

$14 : 21 = \boxed{} : \boxed{}$

$16 : 24 = \boxed{} : \boxed{}$

$18 : 30 = \boxed{} : \boxed{}$

$20 : 25 = \boxed{} : \boxed{}$

$24 : 36 = \boxed{} : \boxed{}$

2 find the common factor

Write a number that divides both parts exactly.

$12 : 18 \quad \text{common factor} = \boxed{}$

$20 : 30 \quad \text{common factor} = \boxed{}$

$24 : 32 \quad \text{common factor} = \boxed{}$

$35 : 49 \quad \text{common factor} = \boxed{}$

3 challenge

Simplify these ratios completely.

$27 : 45 = \boxed{} : \boxed{}$

$32 : 48 = \boxed{} : \boxed{}$

$42 : 56 = \boxed{} : \boxed{}$

$50 : 75 = \boxed{} : \boxed{}$